① $E(f(W)+g(U)) = \iint [f(w)+g(u)]\, p_{UW}(u,w)\, du\, dw$

$= \int f(w) p_W(w)\, dw + \int g(u) p_U(u)\, du$

$= E[f(W)] + E[g(U)]$

② $E[X+a] = \int (x+a) p_X(x)\, dx = E[X] + a\int p_X(x)\, dx$

$= E[X] + a$

③ $E[aX] = \int ax\, p_X(x)\, dx = a E(X)$

④ $Var(aX) = E[(aX - E[aX])^2]$

$= a^2 E[(X - E(X))^2]$

$= a^2 Var(X)$

⑤ $Z = X+Y \Rightarrow Var(Z) = E[(X+Y-E[X]-E[Y])^2]$

$= E[((X-E[X]) + (Y-E[Y]))^2]$

$= Var(X) + Var[Y] + 2C_{XY}$

⑥ $f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$

$= \frac{1}{2\pi\sqrt{ab}} \int_{-\infty}^{\infty} \exp\left[-\frac{1}{2}\left(\frac{x^2}{a} + \frac{(z-x)^2}{b}\right)\right] dx$

$= \frac{a+b}{ab}\left(x - \frac{az}{a+b}\right)^2 + \frac{z^2}{a+b}$

$\Rightarrow f_Z(z) = \frac{e^{-z^2/[2(a+b)]}}{2\pi\sqrt{ab}} \int_{-\infty}^{\infty} \exp\left[-\frac{a+b}{2ab}\left(x - \frac{az}{a+b}\right)^2\right] dx$

$f_Z(z) = \frac{\exp(-z^2/[2(a+b)])}{\sqrt{2\pi(a+b)}}$

(7) $E[X - E[X]] = \int_{-\infty}^{\infty} (x - \mu_x) p_x(x)\, dx$

$= \int x p_x(x)\, dx - \mu_x \int p_x(x)\, dx$

$= \mu_x - \mu_x(1) = 0$

(8) $C_{xy} = E[(X - E[X])(Y - E[Y])]$

$= \iint (x - \mu_x)(y - \mu_y) p_x(x) p_y(y)\, dx\, dy$

$= \left(\int (x - \mu_x) p_x(x)\, dx\right)\left(\int (y - \mu_y) p_y(y)\, dy\right)$

$= 0 \cdot 0 = 0$